

TOPIC 7.A

Structure and diseases of the respiratory system

1. List the upper and lower respiratory tracts. Detail the functioning of the lower respiratory tract.
2. How does the process of inhalation and exhalation take place?
3. What do we mean by vital capacity? How does exercise affect it?
4. List the most common respiratory system diseases (at least three).
5. Your friend smokes regularly. With what arguments would you support advising them to stop smoking?

Expected knowledge	Attainable score	Attained score
1. • Upper: nasal cavity, pharynx. • Lower: larynx, trachea, lungs - within them the two main bronchi, bronchi, bronchioles, alveoli. • Functioning: - larynx: cartilaginous organ, covered by the epiglottis - trachea: C-shaped cartilages, elastic - lungs: main bronchi branch into bronchi, then into bronchioles • bronchioles and alveoli stand in clusters, their surface is simple squamous epithelium, with a rich capillary network, gas exchange occurs on their surface.	1 point 1 point 2 points 4 points	
2. • Inhalation: external intercostal muscles contract, raise the ribs, the chest expands. The dome-shaped diaphragm contracts and flattens. The lungs expand, air pressure decreases, therefore air flows into the lungs. • Exhalation: a passive process. The inspiratory muscles relax, ribs descend, the diaphragm bulges into the thoracic cavity, thus pressure increases, causing air to flow out of the lungs.	3 points 3 points	
3. • The maximal performance of the respiratory organs, the volume of air exhaled with a forced exhalation, is the vital capacity. • Average value: 4-4.8 dm³ - smaller in women than in men • Effect of exercise: inhalation and	2 points 1 point 2 points	

exhalation are more thorough, more alveoli participate in gas exchange, the air capacity of the lungs increases.		
4. • e.g.: - "common cold": of viral origin - influenza (flu): of viral origin - hay fever: a disease of allergic origin - asthma: triggered by the spasmodic contraction of the smooth muscle of the bronchioles and increased mucus production. Triggers: dust mites, pollen, animal dander, etc. - pneumonia: can be caused by bacteria, fungi, viruses - TB (tuberculosis): can be caused by bacterial infection. Inflammatory foci in the lungs. Today there is a vaccine for prevention.	3 points	
5. Logical argumentation is acceptable.	3 points	
6. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 7.B

Characterisation of a plant species

1. Identify the received plant species using the Plant Identification book. Describe the process of identification.
2. To which major taxonomic groups does it belong? What are the characteristics of these groups that can be found on the received species?
3. Characterise the ecological requirements of the species (T-, W-, R-, N-, Z-values).
4. In what type of habitat is the occurrence of this species typical? Where can we find such a place around the school?



Expected knowledge	Attainable score	Attained score
1. Naming the species, listing the identification keys. (Greater celandine - <i>Chelidonium majus</i>)	5 points	
2. Taxonomic group: phylum Angiosperms, class Dicotyledons • pinnate leaf venation, • taproot system, • characteristic flower (4 petals), • pinnately lobed (pinnatifid), crenate, hairy leaf, • silique-like capsule fruit, • its herbaceous stem contains yellow latex.	5 points	
3. Ecological requirements: • T: deciduous forests, W: moderately mesic habitat, R: calcareous soil, N: prefers nitrogen-rich areas, Z: tolerates degradation well.	5 points	
4. Around the school, it occurs everywhere next to the fence in neglected areas. • Medicinal plant: its latex is applied to warts. Internally used for bile and stomach ailments.	5 points	
Total:	20 points	